

NCERT Exemplar Solutions for Class 12

Mathematics

Chapter 12 – Linear Programming

Short Answer Type Question

Example 1: Determine the maximum value of $Z=4x+3y$ if the feasible region for an LPP is shown in Fig. 12.1.

Ans: The feasible region is bounded. So, the maximum of Z must occur at the corner point of the feasible region.

Corner Point	Value of Z
O (0, 0)	$4(0) + 3(0) = 0$
A (25,0)	$4(25) + 3(0) = 100$
B (16, 16)	$4(16) + 3(16) = 112$ (Maximum)
C (0,24)	$4(0) + 3(24) = 72$

So, the maximum value of Z is 112.

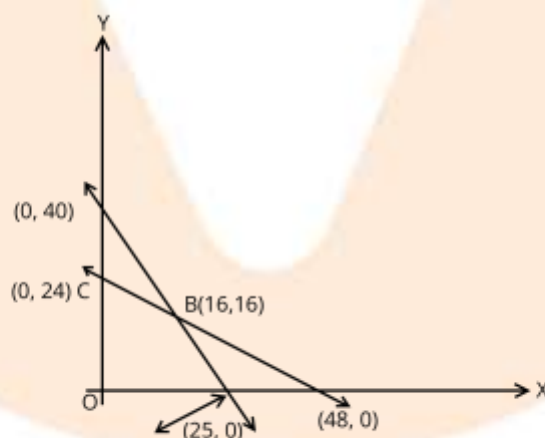


Fig. 12.1

Example 2: Determine the minimum value of $Z=3x + 2y$ (if any), if the feasible region for an LPP is shown in Fig. 12.2.

Ans: The feasible region is unbounded. So the minimum of Z may or may not exist. If it exists, it will be at the corner point(in figure).

Corner Point	Value of Z
A (12, 0)	$3(12)+2(0)=36$
B (4,2)	$3(4)+2(2)=16$
C (1, 5)	$3(1)+2(5)=13$ (smallest)
D (0,10)	$3(0)+2(10)=20$

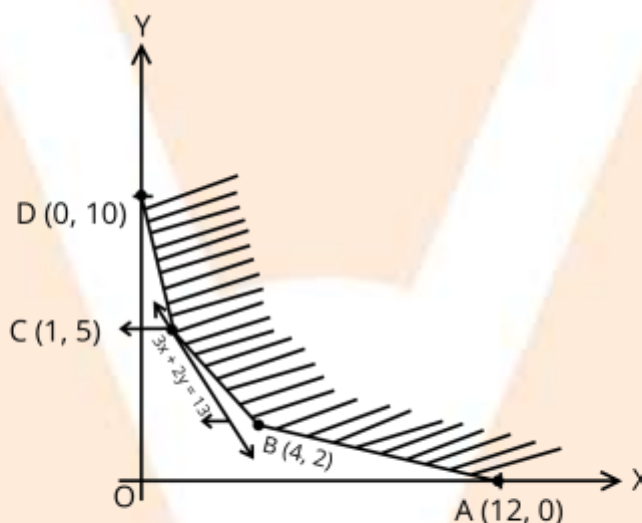


Fig. 12.2

In the graph we see that the open half plane determined by $3x - 2y < 13$ and R do not have a common point. So, the smallest value = 13 is the minimum value of Z .

Example 3: Solve the following LPP graphically:

Maximise $Z=2x+3y$,

Subject to $x + y \leq 4, x \geq 0$

Ans: The shaded region OAB in Fig. is the feasible region determined by the system of constraints $x \geq 0$, $y \geq 0$, $x + y \leq 4$.

The feasible region OAB is bounded, so maximum value will occur at a corner point of the feasible region. Corner Points are O(0, 0), A (4,0) and B (0, 4).

Evaluate Z.

Corner Point	Value of Z
O, (0,0)	$2(0)+3(0)=0$
A (4,0)	$2(4)+3(0)=8$
B (0,4)	$2(0)+3(4)= 12(\text{Maximum})$

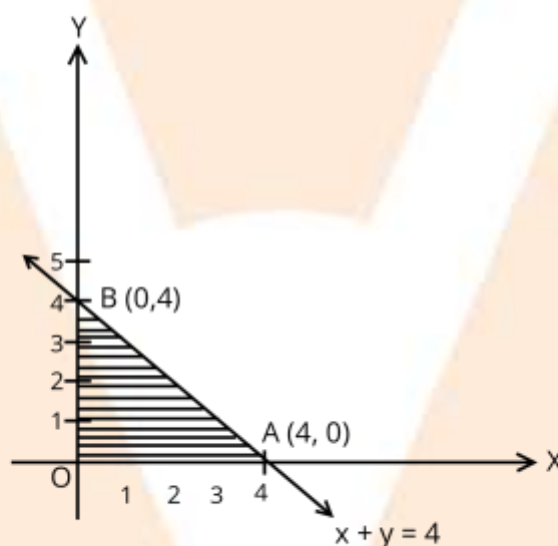


Fig. 12.3

So, the maximum value of $Z = 12$ at the point (0, 4).

Example 4: A manufacturing company makes two types of television sets; one is black and white and the other is colour. The company has resources to make at most 300 sets a week. It takes Rs 1800 to make a black and white set and Rs 2700 to make a coloured set. The company can spend not more than Rs 648000 a week to make television sets. If it makes a profit of Rs 510 per black and white set and Rs 675 per coloured set, how many sets of each type should be produced

so that the company has maximum profit! Formulate this problem as a LPP given that the objective is to maximise the profit.

Ans: Let's assume the number of black and white sets is x and y is the number of coloured sets made each week. Thus

$$x \geq 0, y \geq 0$$

Because the company can make at most 300 sets a week, so, $x + y \leq 300$

Weekly cost (in Rs) of manufacturing the set is

$1800x + 2700y$ and the company can spend upto Rs. 648000. Therefore,

$$1800x + 2700y \leq 648000, \text{ ie., or } 2x + 3y \leq 720$$

The total profit on x black and white sets and y colour sets is Rs($510x - 675y$),
Let $Z = 510x + 675y$. So, this is the objective function.

So, the mathematical formulation of the problem is

Maximise $Z = 510x + 675y$

Subject to the constraints: $x + y \leq 300$

$$2x + 3y \leq 720$$

$$x \geq 0, y \geq 0$$

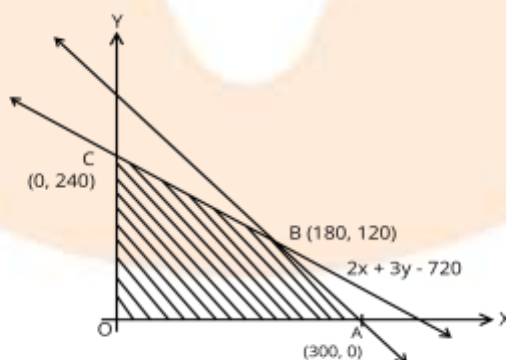


Fig. 12.4

Long Answer Type Question

Example 5: Refer to Example 4, Solve the LPP.

Ans: The problem:

$$\text{Maximise } Z = 510x + 675y$$

$$\text{Subject to the constraints: } x + y \leq 300$$

$$2x + 3y \leq 720$$

$$x \geq 0, y \geq 0$$

The feasible region OABC is shown within the Fig.

Because the feasible region is bounded, a maximum of must occur at the corner point of OBC.

Corner Point	Value of Z
O (0,0)	$510(0) + 675(0) = 0$
A (300, 0)	$510(300) + 675(0) = 153000$
B (180, 120)	$510(180) + 675(120) = 172800$ (maximum)
C (0, 240)	$510(0) + 675(240) = 162000$

So, maximum $Z = 172800$ at the point (180, 120). If the company produces 180 black and white television sets and 120 coloured television sets they will induce maximum profit.

Example 6: Minimise $Z = 3x + 5y$ subject to the constraints:

(A) $x + 2y \geq 10$

(B) $x + y \geq 6$

(C) $3x + y \geq 8$

(d) $x, y \geq 0$

Ans: We first draw the graphs of $x+2y=10$, $x+y=6$, $3x+y=8$. The shaded region ABCD is the feasible region determined by the above restriction. The feasible region is unbounded. Therefore, a minimum of 7. may or may not occur. If it happens, it will be on the corner point.

Corner Point	Value of
A (0,8)	40
B (1,5)	28
C (2,4)	26(smallest)
D (10,0)	30

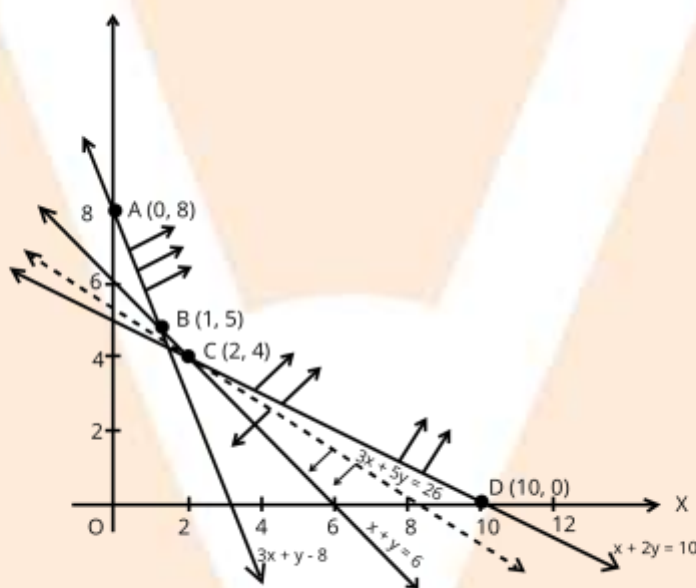


Fig. 12.5

In the graph of $3x + 5y \leq 26$ as shown in Fig. by dotted line. We see that the open half plane determined by $3x + 5y \leq 26$ and R do not have a point in common. Thus, the minimum value of Z is 26.

Objective Type Questions

Choose the correct answer from the given four options in each of the Examples 7 to 8.

Example 7: The corner points of the feasible region determined by the system of linear constraints are $(0, 10)$, $(5, 5)$, $(15, 15)$, $(0, 20)$. Let $Z = px + qy$, where $p, q > 0$. Condition on p and q so that the maximum of Z occurs at both the points $(15, 15)$ and $(0, 20)$ is

- (A) $p=q$
- (B) $p=2q$
- (C) $q=2p$
- (D) $q=3p$

Ans: Correct option - D

Because Z occurs maximum at $(15, 15)$ and $(0, 20)$

Therefore, $15p + 15q = 0.p + 20.q$

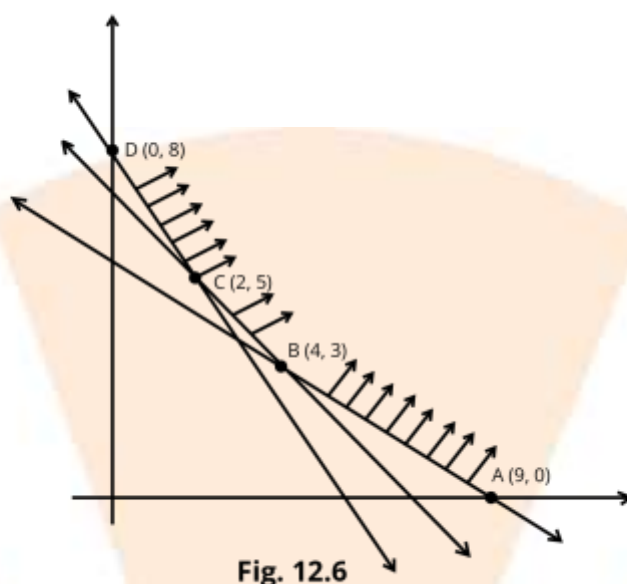
$$15p + 15q = 20q$$

$$15p = 5q$$

$$\text{or, } q = 3p$$

Example 8: Feasible region (shaded) for a LPP is shown in the Fig. Minimum of $Z = 4x + 3y$ occurs at the point

- (A) $(0,8)$
- (B) $(2,5)$
- (C) $(4,3)$
- (D) $(9,0)$



Ans: Correct option - B

Fill in the blanks in each of the Examples 9 and 10.

Example 9: In a LPP, the linear function which has to be maximised or minimised is called a linear..... function.

Ans: In a LPP, the linear function which has to be maximised or minimised is called a linear Objective function.

Example 10: The common region determined by all the linear constraints of a LPP is called the..... region.

Ans: The common region determined by all the linear constraints of a LPP is called the feasible region.

State whether the statements in Examples 11 and 12 are True or False.

Example 11: If the feasible region for a linear programming problem is bounded, then

The objective function $Z = ax + by$ has both a maximum and a minimum value on R.

Ans: True

Example 12: The minimum value of the objective function $Z = ax + by$ in a linear programming problems always occur at only one corner point of the feasible region.

Ans: False

Because the minimum value can also occur at more than one corner points of the feasible region.

Exercise

Short Answer Questions

1. Determine the maximum value of $Z = 11x + 7y$ subject to the constraints:

$$2x + y \leq 6, x \leq 2, x \geq 0, y \geq 0.$$

Ans: We have to maximise $Z = 11x + 7y$ (i)

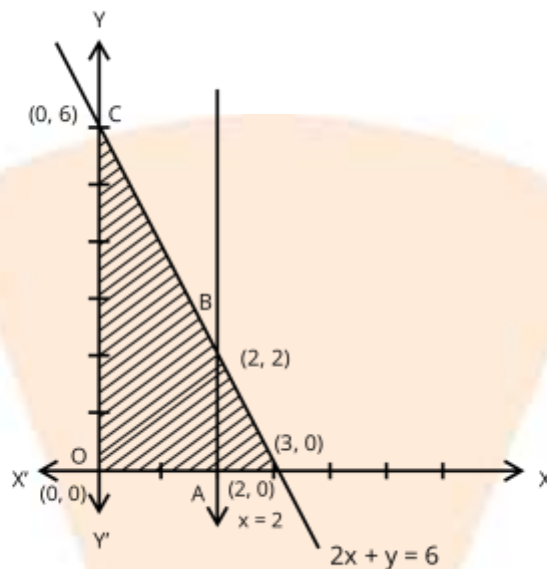
Subject to the constraints

$$2x + y \leq 6 \dots (ii)$$

$$x \leq 2 \dots (iii)$$

$$x \geq 0, y \geq 0 \dots (iv)$$

We plotted these inequalities shown in the figure.



The shaded region is shown as OABC is bounded and the corner points are $O(0,0)$, $A(2,0)$, $B(2,2)$ and $C(0,6)$.

Corner Point	Value of Z
$(0,0)$	0
$(2,0)$	22
$(2,2)$	36
$(0,6)$	42 (maximum)

So, the maximum value of Z is 42 at $(0, 6)$.

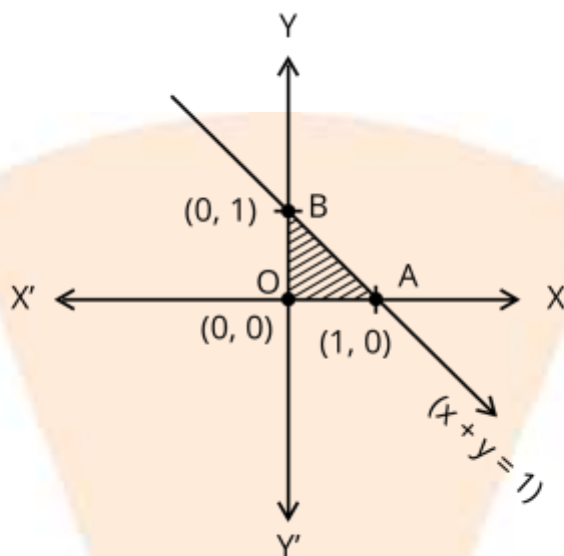
2. Maximise $Z=3x+4y$, subject to the constraints: $x + y \leq 1, x \geq 0, y \geq 0$.

Ans: Maximise $Z = 3x+4y$

Subject to the constraints

$$x + y \leq 1, x \geq 0, y \geq 0$$

We plotted these inequalities shown in the figure.



The Shaded region shown in the figure as OAB is bounded and the corner points are $O(0,0)$, $A(1,0)$, $B(0,1)$.

Corner Point	Value of Z
$(0,0)$	0
$(1,0)$	3
$(0,1)$	4 (maximum)

So, the maximum value of Z is 4 at $(0, 1)$.

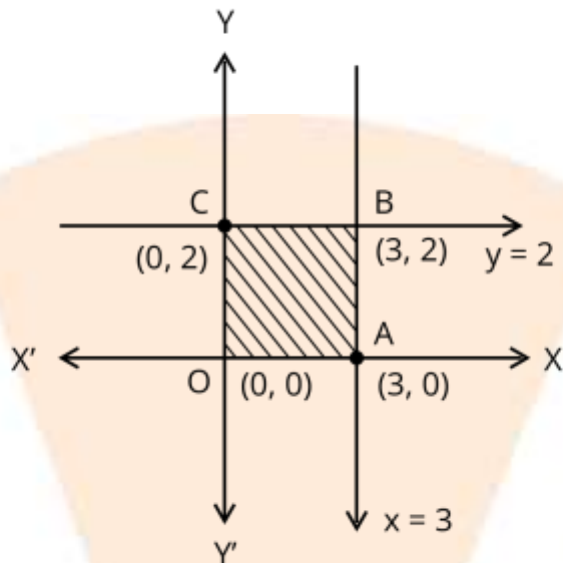
3. Maximise the function $Z=11x+7y$, subject to the constraints: $x \leq 3$, $y \leq 2$, $x \geq 0$, $y \geq 0$.

Ans: Maximise $Z=11x+7y$

Subject to the constraints

$$x \leq 3, y \leq 2, x \geq 0, y \geq 0.$$

We plotted these inequalities shown in the figure.



Shaded region shown in the figure as OABC is bounded and the corner points are $O(0,0)$, $A(3,0)$, $B(3,2)$ and $C(0,2)$.

Corner Point	Value of Z
$(0,0)$	0
$(3,0)$	33
$(3,2)$	47 (maximum)
$(0,2)$	14

So, the maximum value of is 47 at the point $(3, 2)$.

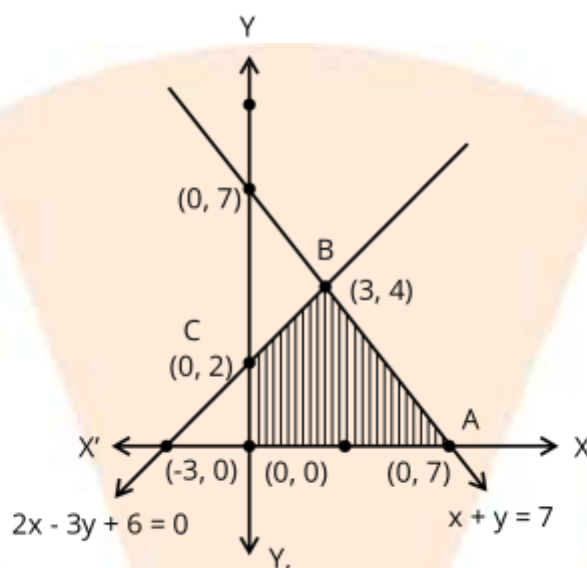
4. Minimise $Z = 13x - 15y$ subject to the constraints: $x + y \leq 7$, $2x - 3y + 6 \geq 0$, $x \geq 0, y \geq 0$.

Ans: Minimise $Z = 13x - 15y$

Subject to the constraints:

$$x + y \leq 7, 2x - 3y + 6 \geq 0, x \geq 0, y \geq 0.$$

We plotted these inequalities shown in the figure.



Shaded region shown as OABC is bounded and corner points are O(0,0), A(7,0), B(3,4) and C(0, 2).

Corner Point	Value of
(0,0)	0
(7,0)	91
(3,4)	-21
(0,2)	-30 (minimum)

So, the minimum value of is -30 at (0,2).

5. Determine the maximum value of $Z = 3x + 4y$ if the feasible region (shaded) for a LPP is shown in Fig.

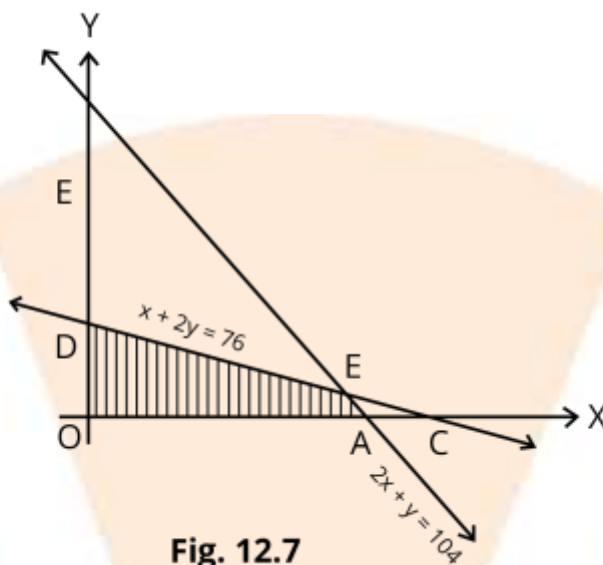


Fig. 12.7

Ans: In given graph lines $x + 2y = 76$ and $2x + y = 104$ intersect at E(44,16)

As clear from the graph, corner points are O. A. E and D with coordinates (0,0), (52,0), (44,16) and (0,38), respectively.

Also, a given region is bounded.

$$Z = 3x + 4y$$

Corner Point	Value of
(0,0)	0
(52,0)	156
(44,16)	196 (maximum)
(0,38)	152

So, it is maximum at (44,16) and its maximum value is 196.

6. Feasible region (shaded) for a LPP is shown in Fig. Maximise $Z = 5x + 7y$.

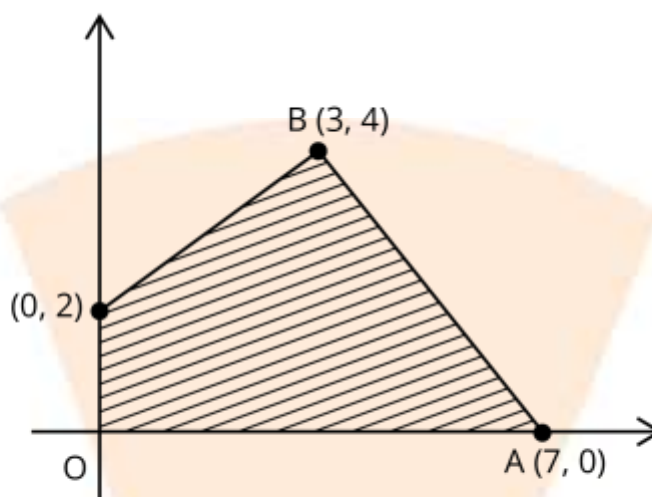


Fig. 12.8

Ans: The Shaded region is bounded and coordinate of corner points $O(0,0)$, $A(7,0)$, $B(3,4)$ and $C(0,2)$.

Also, $Z = 5x + 7y$.

Corner Point	Value of
$(0,0)$	0
$(7,0)$	35
$(3,4)$	43 (maximum)
$(0,2)$	14

So, the maximum value of is 43 at $(3,4)$.

7. The feasible region for a LPP is shown in Fig. Find the minimum value of $Z = 11x + 7y$.

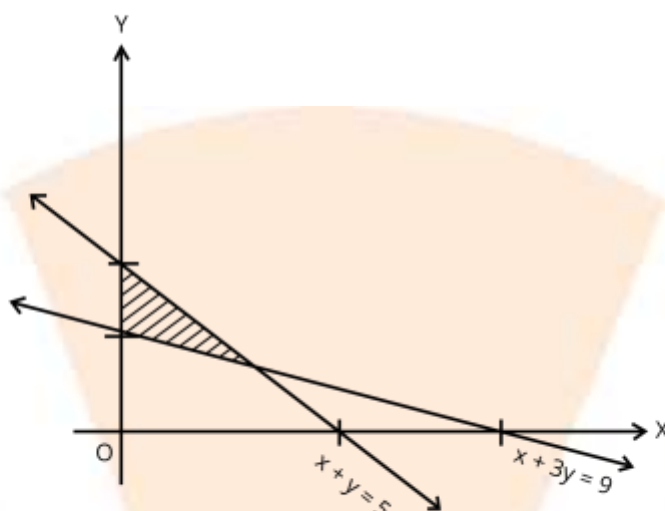


Fig. 12.9

Ans: From the figure, the feasible region is bounded with coordinates of corner points as C(0,3), A(3,2) and B(0,5).

Also, $Z = 11x + 7y$.

Corner Point	Value of
(0,3)	21 (Minimum)
(3,2)	47
(0,5)	35

So, the minimum value of Z is 21 at (0,3).

8. Refer to Exercise 7 above. Find the maximum value of.

Ans: It is clear that, it is maximum at (3, 2) and its maximum value is 47.

9. The feasible region for a LPP is shown in Fig. Evaluate $Z = 4x + y$ at each of the corner points of this region. Find the minimum value of, if it exists.

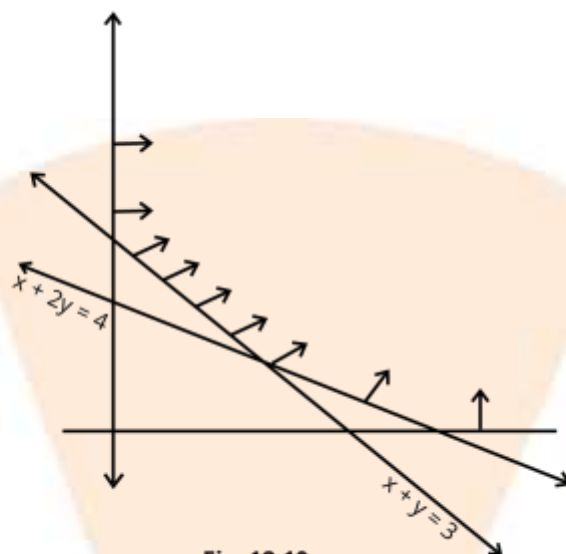
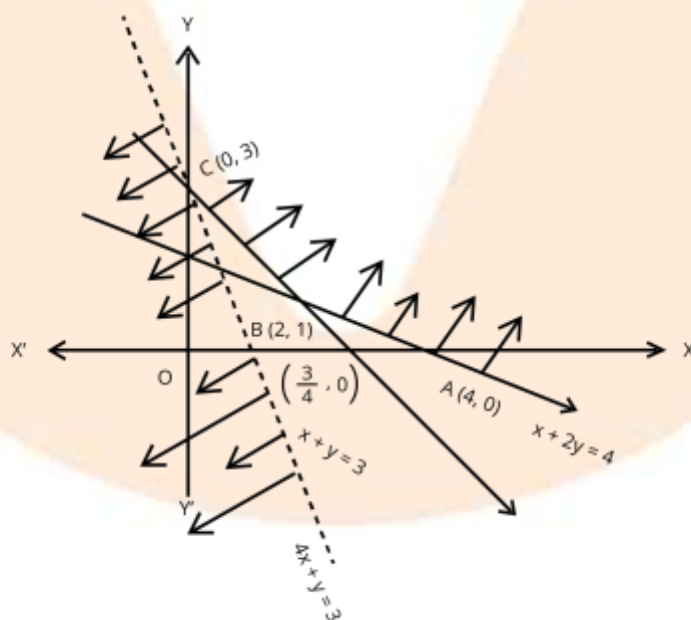


Fig. 12.10

Ans: From the shaded region, the feasible region is unbounded with the corner points $A(4,0)$, $B(2,1)$ and $C(0,3)$.

Also, $Z = 4x + y$.



Corner Point	Value of
(4,0)	16
(2,1)	9
(0,3)	3 (Maximum)

And now, we see that 3 is the smallest value of at the corner point (0, 3). Here we see that the region is unbounded, so 3 may or may not be the minimum value. To decide this issue, we graph the inequality $4x + y \leq 3$ and check whether the resulting open half plane has no point in common with the feasible region; otherwise, has no minimum common value.

From the graph, it is clear that there is no point in common with the region and hence has a minimum value of 3 at (0, 3).

10. In Fig. 12.11, the feasible region (shaded) for a LPP is shown. Determine the maximum and minimum value of $Z = x + 2y$.

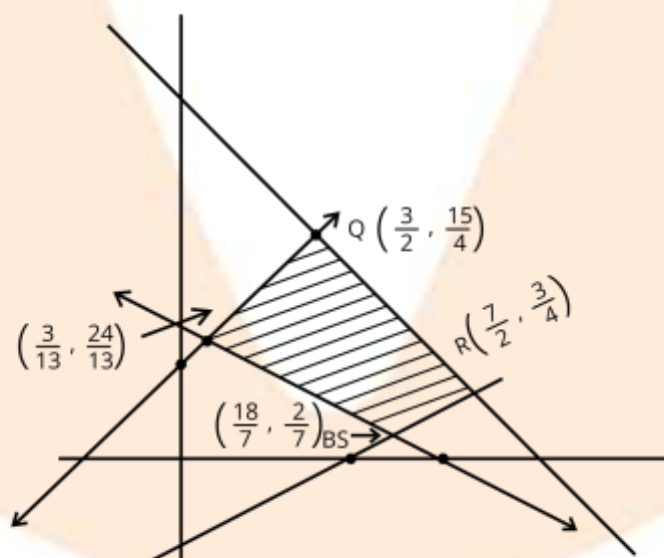


Fig. 12.11

Ans: From the shaded bounded region, the coordinates of corner points are $P(\frac{3}{13}, \frac{24}{13})$, $Q(\frac{3}{2}, \frac{15}{4})$, $R(\frac{7}{2}, \frac{3}{4})$ and $S(\frac{18}{7}, \frac{2}{7})$

Also, $Z = x + 2y$.

Corner Point	Value of
$(\frac{3}{13}, \frac{24}{13})$	$\frac{51}{13} = 3\frac{12}{13}$
$(\frac{18}{7}, \frac{2}{7})$	$\frac{22}{7} = 3\frac{1}{7}$ (minimum)
$(\frac{7}{2}, \frac{3}{4})$	$\frac{20}{4} = 5$
$(\frac{3}{2}, \frac{15}{4})$	$\frac{36}{4} = 9$ (maximum)

So, the maximum and minimum value of are 9 and $3\frac{1}{7}$ respectively.

11. A manufacturer of electronic circuits has a stock of 200 resistors, 120 transistors and 150 capacitors and is required to produce two types of circuits A and B. Type A requires 20 resistors, 10 transistors and 10 capacitors. Type B requires 10 resistors, 20 transistors and 30 capacitors. If the profit on type A circuit is Rs 50 and that on type B circuit is Rs 60, formulate this problem as a LPP so that the manufacturer can maximise his profit.

Ans: Let the manufacturer produce x units of type A circuit and y units of type B circuits. From the given information.

We have the following corresponding constraint table.

	Type A(X)	Type B(y)	Maximum stock
Resistors	20	10	200
Transistors	10	20	120

Capacitors	10	30	150
Profit	Rs 50	Rs 60	

Thus, total profit $Z=50x+60y$ (in Rs).

Now, we have the following mathematical model for the given problem.

Maximise $Z=50x+60y$ (i)

Subject to the constraints.

$20x + 10y \leq 200$ [resistor constraint]

$\Rightarrow 2x + y \leq 20$ (ii)

And $10x + 20y \leq 120$ [transistor constraint]

$\Rightarrow x + 2y \leq 12$ (iii)

And $10x + 30y \leq 150$ [capacitor constraint]

$\Rightarrow x + 3y \leq 15$...(iv)

And $x \geq 0, y \geq 0$ (v) [non-negative constraint]

So, Maximise $Z = 50x + 60y$, subject to $2x + y \leq 20, x + 2y \leq 12, x + 3y \leq 15, x \geq 0, y \geq 0$.

12. A firm has to transport 1200 packages using large vans which can carry 200 packages each and small vans which can take 80 packages each. The cost for engaging each large van is Rs 400 and each small van is Rs 200. Not more than Rs 3000 is to be spent on the job and the number of large vans cannot exceed the number of small vans. Formulate this problem as a LPP given that the objective is to minimise cost. job and the number of large vans cannot exceed the number of small vans. Formulate this problem as a LPP given that the objective is to minimise cost.

Ans: Let the firm have a x number of large vans and y number of small vans. From the given information, we have the following corresponding constraint table.

	Large vans(X)	Small vans(y)	Maximum/Minimum
Package	200	80	1200
Cost	400	200	3000

Thus, we see that the objective function for minimum cost is $Z=400x+200y$

Subject to constraints

$$200x + 80y \geq 1200 \quad [\text{package constraint}]$$

$$\Rightarrow 5x + 2y \geq 30 \quad \dots\dots(i)$$

$$\text{And } 400x + 200y \leq 3000 \quad [\text{cost constraint}]$$

$$\Rightarrow 2x + y \leq 15 \quad \dots(ii)$$

$$\text{And } x \leq y \quad \dots(iii)$$

$$\text{And } x \geq 0, y \geq 0 \quad [\text{non-negative constraints}]\dots(iv)$$

Thus, required LPP to minimise cost is minimise $Z = 400x + 200y$, subject to $5x + 2y \geq 30, 2x + y \leq 15, x \leq y, x \geq 0, y \geq 0$.

13. A company manufactures two types of screws A and B. All the screws have to pass through a threading machine and a slotting machine. A box of Type A screws requires 2 minutes on the threading machine and 3 minutes on the slotting machine. A box of type B screws requires 8 minutes of threading on the threading machine and 2 minutes on the slotting machine. In a week, each machine is available for 60 hours.

On selling these screws, the company gets a profit of Rs 100 per box on type A screws and Rs 170 per box on type B screws. Formulate this problem as a LPP given that the objective is to maximise profit.

Ans: Let the company manufacture x boxes of type A screws and y boxes of type B screws. From the given information, we have following corresponding constraint table

	Type A(x)	Type B(y)	Maximum time available on each machine in a week
Time required for screws on threading machine	200	80	60x60 (min)
Time required for screws on slotting machine			60x60 (min)
Profit	Rs 100	Rs 170	

Thus, we see that the objective function for maximum profit is $Z = 100x + 170y$.

Subject to constraints.

$$2x + 8y \leq 60 \times 60 \quad [\text{time constraint for threading machine}]$$

$$\Rightarrow x + 4y \leq 1800 \quad \dots\dots(i)$$

$$\text{And } 3x + 2y \leq 60 \times 60 \quad [\text{time constraint for slotting machine}]$$

$$\Rightarrow 3x + 2y \leq 3600 \quad \dots\dots(ii)$$

$$\text{Also, } x \geq 0, y \geq 0 \quad [\text{non-negative constraints}]\dots(iii)$$

Required LPP is,

Maximise $Z = 100x + 170y$, Subject to constraints $x + 4y \leq 1800, 3x + 2y \leq 3600, x \geq 0, y \geq 0$.

14. A company manufactures two types of sweaters: type A and type B. It costs Rs 360 to make a type A sweater and Rs 120 to make a type B sweater. The company can make at most 300 sweaters and spend at most Rs 72000 a day. The number of sweaters of type B cannot exceed the number of sweaters of type A by more than 100. The company makes a profit of Rs 200 for each sweater of

type A and Rs 120 for every sweater of type B. Formulate this problem as a LPP to maximise the profit of the company.

Ans: Let the company manufacture x number of type A sweaters and y number of type B sweaters. The company spent at most Rs 72000 a day.

$$360x + 120y \leq 72000$$

$$\Rightarrow 3x + y \leq 600 \dots (i)$$

Also, the company can make at most 300 sweaters.

$$x + y \leq 300 \dots (ii)$$

Further, the number of sweaters of type B cannot exceed the number of sweaters of type A by more than 100.

$$y - x \leq 100 \dots (iii)$$

And $x \geq 0, y \geq 0 \dots (iv)$

Thus, the required LPP to maximise the profit is Maximise $Z = 200x + 120y$ is subject to constraints.

$$3x + y \leq 600$$

$$x + y \leq 300$$

$$x - y \geq -100$$

$$x \geq 0, y \geq 0$$

15. A man rides his motorcycle at the speed of 50 km/hour. He has to spend Rs 2 per km on petrol. If he rides it at a faster speed of 80 km/hour, the petrol cost increases to Rs 3 per km. He has at most Rs 120 to spend on petrol and one hour time. He wishes to find the maximum distance that he can travel. Express this problem as a linear programming problem.

Ans: Let the man ride his motorcycle to a distance x km at the speed of 50 km/h and to a distance y km at the speed of 80 km/h.

Therefore, the cost of petrol is $2x + 3y$.

Since, he has to spend Rs 120 at most on petrol.

$$2x + 3y \leq 120 \dots (i)$$

Also, he has at most 1 hour time.

$$\frac{x}{50} + \frac{y}{80} \leq 1$$

$$\Rightarrow 8x + 5y \leq 400 \dots (ii)$$

Also, $x \geq 0$, $y \geq 0 \dots (iii)$ [non-negative constraints]

Thus, required LPP to travel maximum distance by him is

Maximise $Z = x + y$, subject to constraints $2x + 3y \leq 120, 8x + 5y \leq 400, x \geq 0, y \geq 0$.

Long Answer Questions

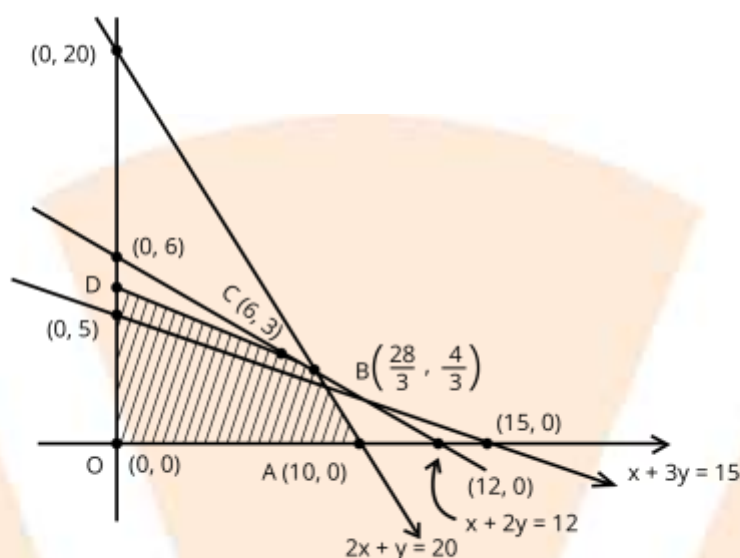
16. Refer to Exercise 11. How many of circuits of Type A and of Type B, should be produced by the manufacturer so as to maximise his profit? Determine the maximum profit.

Ans: Referring to solution 11, we have

Maximise $Z = 50x + 60y$, subject to

$$2x + y \leq 20, x + 2y \leq 12, x + 3y \leq 15, x \geq 0, y \geq 0$$

We plotted these inequalities shown in the figure.



From the shaded region OABCD is bounded with corner points $O(0,0)$, $A(10,0)$, $B(\frac{28}{3}, \frac{4}{3})$, $C(6,3)$ and $D(0,5)$.

Corner Point	Value of
$(0,0)$	0
$(10,0)$	500
$(\frac{28}{3}, \frac{4}{3})$	546.66 (Maximum)
$(6,3)$	480
$(0,5)$	300

So the required maximum profit is 480 where circuits of type A is 6 and circuits of type B is 3.

17. Refer to Exercise 12. What will be the minimum cost?

Ans: Referring to solution 12, we have

Minimise $Z = 400x + 200y$, subject to

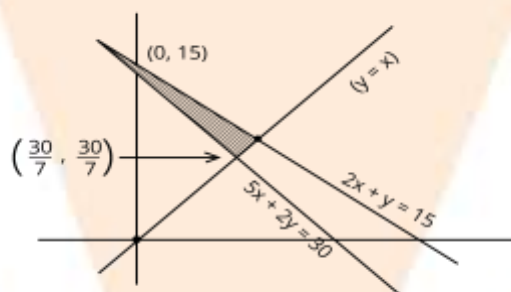
$$5x + 2y \geq 30$$

$$2x + y \leq 15$$

$$x \leq y$$

$$x \geq 0, y \geq 0$$

We plotted these inequalities shown in the figure.



The shaded feasible region is bounded and corner points are $(\frac{30}{7}, \frac{30}{7})$, $(0, 15)$ and $(5, 5)$.

Corner Point	Value of
$(0, 15)$	3000
$(5, 5)$	3000
$(\frac{30}{7}, \frac{30}{7})$	2571.43 (minimum)

So, the minimum cost is Rs 2571.43.

18. Refer to Exercise 13. Solve the linear programming problem and determine the maximum profit to the manufacturer.

Ans: Referring to solution 13, we have

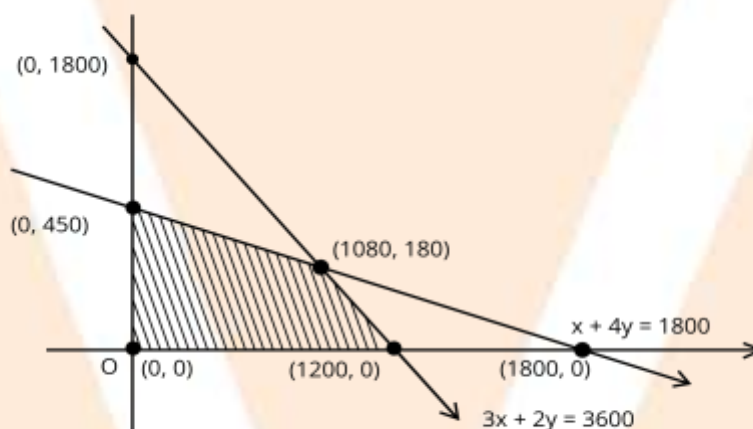
Maximise $Z = 100x + 170y$, subject to

$$3x + 2y \leq 3600$$

$$x + 4y \leq 1800$$

$$x \geq 0, y \geq 0$$

We plotted these inequalities shown in the figure.



The shaded region is bounded and corner points are (0,0), (1200,0), (1080,180) and (0,450).

Corner Point	Value of
(0,0)	0
(1200,0)	120000
(1080,180)	138600 (Maximum)
(0,450)	46500

So, the maximum profit to the manufacturer is 138600.

19. Refer to Exercise 14. How many sweaters of each type should the company day to get a maximum profit? What is the maximum profit?

Ans: Referring to solution 14, we have

Maximise $Z = 200x + 120y$, Subject to

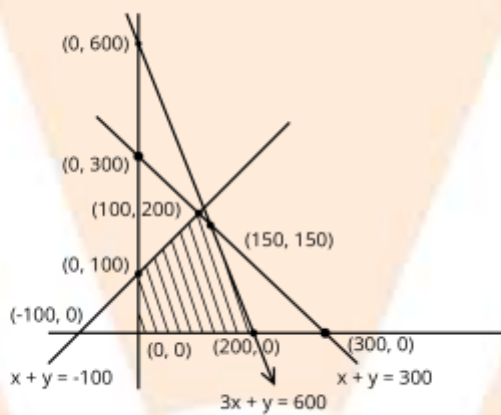
$$x + y \leq 300$$

$$3x + y \leq 600$$

$$x - y \geq -100$$

$$x \geq 0, y \geq 0$$

We plotted these inequalities shown in the figure.



The shaded feasible region is bounded and corner points are:

(0,0), (200,0), (150,150), (100,200) and (0,100).

Corner Point	Value of
(0,0)	0
(200,0)	40000
(150,150)	48000 (Maximum)
(100,200)	44000
(0,100)	12000

So, 150 sweaters of each type are made by the company and maximum profit is Rs. 48000.

20. Refer to Exercise 15. Determine the maximum distance that the man can travel.

Ans: Referring to solution 15, we have

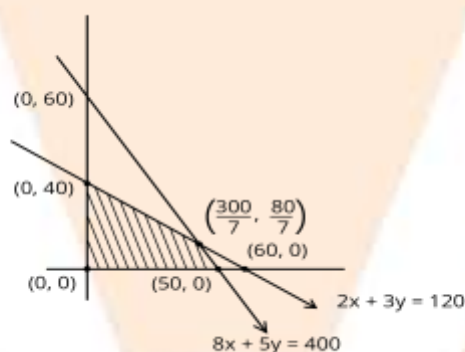
Maximise $Z = x + y$, subject to

$$2x + 3y \leq 120$$

$$8x + 5y \leq 400$$

$$x \geq 0, y \geq 0$$

We plotted these inequalities shown in the figure.



The shaded feasible region is bounded and corner points are $(0, 0)$, $(50, 0)$, $(\frac{300}{7}, \frac{80}{7})$ and $(0, 40)$.

Corner Point	Value of
$(0,0)$	0
$(50,0)$	50
$(\frac{300}{7}, \frac{80}{7})$	$54\frac{2}{7}$ (Maximum)
$(0,40)$	40

So, the maximum distance that the man can travel is $54\frac{2}{7}$ km.

21. Maximise $Z = x + y$, subject to $x + 4y \leq 8$, $2x + 3y \leq 12$, $3x + y \leq 9$, $x \geq 0$, $y \geq 0$.

Ans: We have

Maximise $Z = x + y$, subject to,

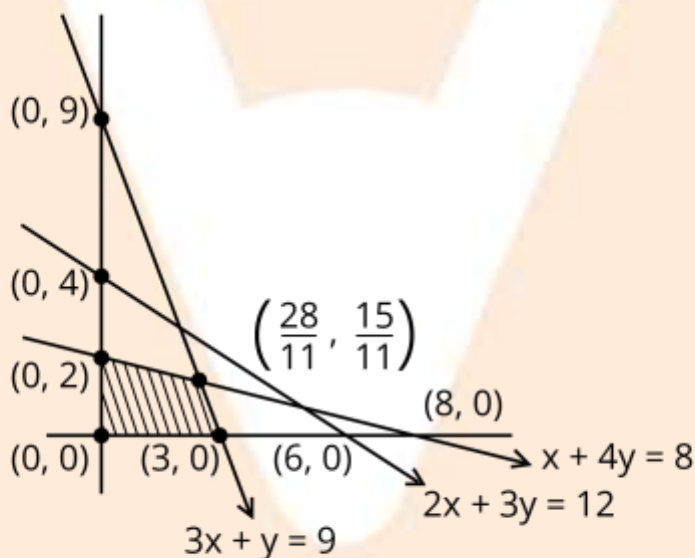
$$x + 4y \leq 8$$

$$2x + 3y \leq 12$$

$$3x + y \leq 9$$

$$x \geq 0, y \geq 0.$$

We plotted these inequalities shown in the figure.



The shaded feasible region is bounded and corner points are $(0, 0)$, $(3, 0)$, $(\frac{28}{11}, \frac{15}{11})$ and $(0, 2)$.

Corner Point	Value of
--------------	----------

(0,0)	0
(3,0)	3
$(\frac{28}{11}, \frac{15}{11})$	$3\frac{10}{11}$ (Maximum)
(0,2)	2

So, the maximum value is $3\frac{10}{11}$.

22. A manufacturer produces two models of bikes - model X and model Y. Model X takes 6 man-hours to make per unit, while model Y takes 10 man-hours per unit. There are a total of 450 man-hours available per week. Handling and marketing costs are Rs 2000 and Rs 1000 per unit for models X and Y respectively. The total funds available for these purposes are Rs 80,000 per week. Profits per unit for models X and Y are Rs 1000 and Rs 500, respectively. How many bikes of each model should the manufacturer produce so as to yield a maximum profit? Find the maximum profit.

Ans: Let x number of models x and y number of model y bikes.

Given, model x takes 6 man-hours to make per unit and model Y takes 10 man-hours to make per unit.

Given, there are a total of 450 man-hours available per week.

$$\text{So, } 6x + 10y \leq 450$$

$$\Rightarrow 3x + 5y \leq 225 \dots (i)$$

Also, for model X and Y handling and marketing costs are Rs 2000 and Rs 1000, respectively and total funds available for these purposes are Rs 80000 per week.

$$2000x + 1000y \leq 80000$$

$$\Rightarrow 2x + y \leq 80 \dots (ii)$$

Also, $x \geq 0$, $y \geq 0$.

So, the profits per unit for models x and y are Rs 1000 and Rs 500, respectively,

We have

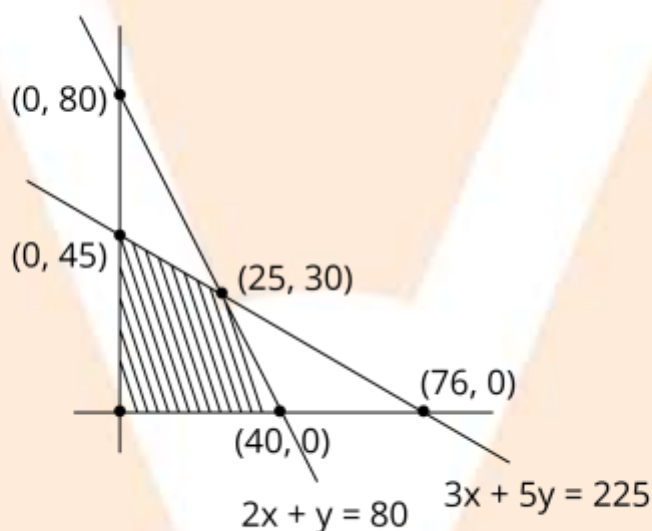
Maximise $Z = 1000x + 500y$, Subject to

$$3x + 5y \leq 225$$

$$2x + y \leq 80$$

$$x \geq 0, y \geq 0$$

We plotted these inequalities shown in the figure.



The shaded feasible region is bounded and corner points are (0,0), (40,0), (25,30) and (0,45).

Corner Point	Value of
(0,0)	0
(40,0)	40000 (maximum)
(25,30)	40000(maximum)
(0,45)	22500

So, to get maximum profit the manufacturer should produce 25 bikes of model X and 30 bikes.

23. In order to supplement a daily diet, a person wishes to take some X and some Y tablets. The contents of iron, calcium and vitamins in X and Y (in milligram per tablet) are given as below:

Tablets	Iron	Calcium	Vitamin
x	6	3	2
y	2	3	4

The person needs at least 18 milligram of iron, 21 milligram of calcium and 16 milligram of vitamins. The price of each tablet of X and Y is Rs 2 and Re 1 respectively. How many tablets of each should the person take in order to satisfy the above requirement at the minimum cost?

Ans.: Let take x units of tablet x and y unit of tablet y.

We have

$$6x + 2y \geq 18$$

$$\Rightarrow 3x + y \geq 9 \dots (i)$$

$$3x + 3y \geq 21$$

$$\Rightarrow x + y \geq 7 \dots (ii)$$

And $2x + 4y \geq 16$

$$\Rightarrow x + 2y \geq 8 \dots (iii)$$

Also, $x \geq 0, y \geq 0 \dots (iv)$

Given the price of each tablet of x and y is Rs 2 and Rs 1 respectively.

So,

Minimise $Z = 2x + y$, subject to

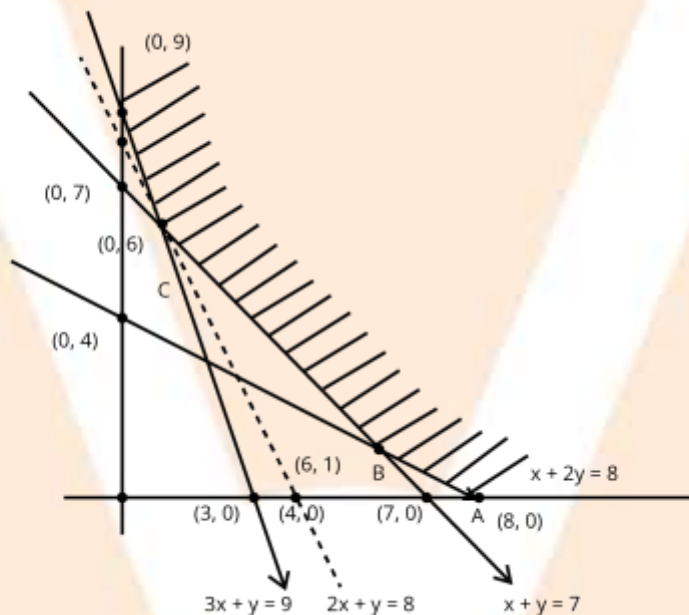
$$3x + y \geq 9$$

$$x + y \geq 7$$

$$x + 2y \geq 8$$

$$x \geq 0, y \geq 0$$

We plotted these inequalities shown in the figure.



The shaded feasible region is bounded and corner points are (8,0), (6,1), (1,6) and (0,9).

Corner Point	Value of
(8,0)	16
(6,1)	13
(1,6)	8(minimum)
(0,9)	9

Here, we see that 8 is the minimum value of at (1,6) but the feasible region is unbounded. So, 8 may or may not be the minimum value.

To decide, we graph the inequality $2x + y \leq 8$ and check whether it has a common point.

We see from the graph that there is no common point.

So, the person should take 1 unit of x tablet and 6 units of y tablets to satisfy requirements and at the minimum cost of Rs 8.

24. A company makes 3 models of calculators: A, B and C at factory I and factory II. The company has ordered for at least 6400 calculators of model A, 4000 calculators of model B and 4800 calculators of model C. At factory 1, 50 calculators of model A, 50 of model B and 30 of model C are made every day; at factory II, 40 calculators of model A. 20 of model B and 40 of model C are made everyday. It costs Rs 12000 and Rs 15000 each. day to operate factory I and II, respectively. Find the number of days each factory should operate to minimise the operating costs and still meet the demand.

Ans: Let factory I be operated for x days and II operate for y days.

At factory I, 50 calculators of model A and at factory II, 40 calculators of model A are made everyday.

Given, the company has ordered at least 6400 calculators of model A.

$$50x + 40y \geq 6400$$

$$\Rightarrow 5x + 4y \geq 640 \dots (i)$$

Also, at factory I, 50 calculators of model B and at factory II, 20 calculators of model B are made everyday.

Since then, the company has ordered at least 4000 calculators of model B.

$$50x + 20y \geq 4000$$

$$\Rightarrow 5x + 2y \geq 400 \dots (ii)$$

Similarly, for model C

$$30x + 40y \geq 4800$$

$$\Rightarrow 3x + 4y \geq 480 \dots(\text{iii})$$

$$\text{Also, } x \geq 0, y \geq 0 \dots (\text{iv})$$

It costs Rs 12000 and Rs 15000 to operate factories I and II, respectively.

Minimise $Z = 12000x + 15000y$, subject to

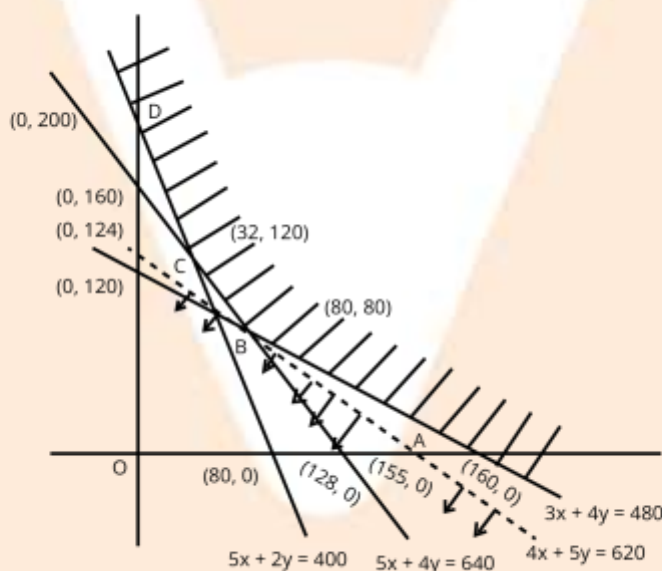
$$5x + 4y \geq 640$$

$$5x + 2y \geq 400$$

$$3x + 4y \geq 480.$$

$$x \geq 0, y \geq 0$$

We plotted these inequalities shown in the figure.



The shaded feasible region is bounded and corner points are (160,0), (80,60), (32,120) and (0,200).

Corner Point	Value of
--------------	----------

(160,0)	1920000
(80,60)	1860000 (minimum)
(32,120)	2184000
(0,200)	3000000

We see from the graph that there is no common point.

So, it has no common points, so $Z = 12000x + 15000y$ has a minimum value of 1860000 at (80,60).

Factory I; 80 days

Factory II; 60 days.

25. Maximise and Minimise $Z=3x-4y$, subject to

(A) $x-2y \leq 0$

(B) $-3x+y \leq 4$

(C) $x-y \leq 6$

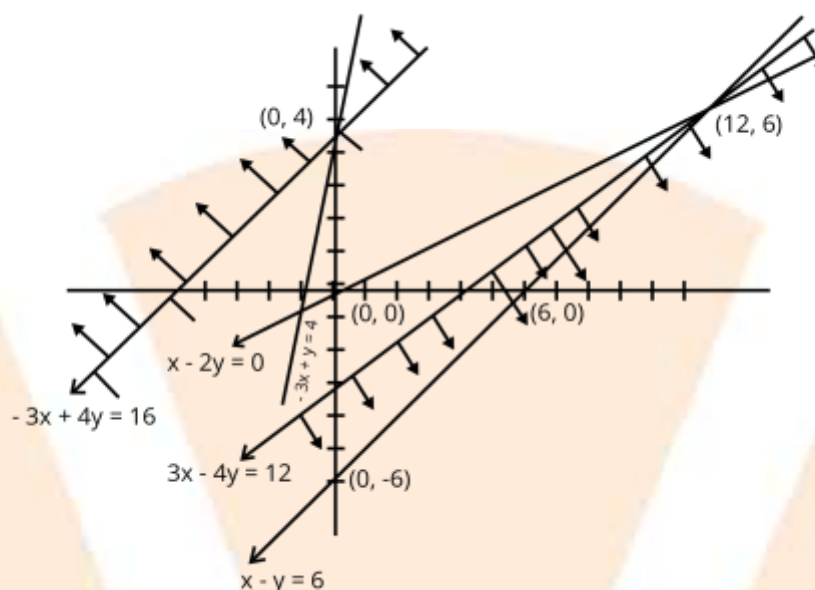
(D) $x, y \geq 0$

Ans: We have

Maximise and minimise $Z=3x-4y$, subject to

$x-2y \leq 0, -3x+y \leq 4, x-y \leq 6, x, y \geq 0$

We plotted these inequalities shown in the figure.



The shaded feasible region is bounded and corner points are $(0,0)$, $(12,6)$ and $(0,4)$.

Corner Point	Value of
$(0,0)$	0
$(0,4)$	-16
$(12,6)$	12

For this given unbounded region, the value of may or may not be -16. So.

To decide, we graph the inequality $3x - 4y \leq -16$ and check whether it has a common point.

We see from the graph that there is a common point. So, it does not have any minimised value.

Also, similarly for maximum value, we graph the inequality $3x - 4y \geq 12$

We see from the graph that there is no common point.

So, the maximum value is 12.

Objective Type Questions

Choose the correct answer from the given four options in each of the Exercises 26 to 34.

26. The corner points of the feasible region determined by the system of linear constraints are (0,0), (0,40), (20,40), (60,20), (60,0). The objective function is $Z=4x+3y$.

Compare the quantity in Column A and Column B

Column A	Column B
Maximum of Z	325

(A) The quantity in column A is greater

(B) The quantity in column B is greater

(C) The two quantities are equal

(D) The relationship cannot be determined on the basis of the information Supplied

Ans: Correct option - B

Corner points	Value of
(0,0)	0
(0,40)	120
(20,40)	200
(60,20)	300 (maximum)
(60,0)	240

So, maximum value of $Z=300<325$

27. The feasible solution for a LPP is shown in Fig. Let $Z=3x-4y$ be the objective function. Minimum of Z occurs at

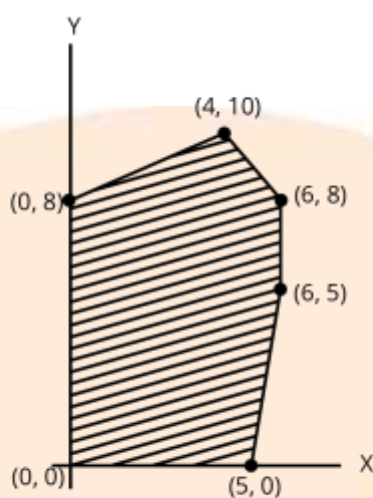


Fig. 12.12

(A) (0, 0)

(B) (0,8)

(C) (5,0)

(D) (4, 10)

Ans: Correct option - B

Corner points	Value of
(0,0)	0
(5,0)	15 (maximum)
(6,5)	-2
(6,8)	-14
(4,10)	-28
(0,8)	-32 (minimum)

So, the minimum of Z occurs at (0, 8).

28. Refer to Exercise 27. Maximum of Z occurs at

- (A) (5,0)
- (B) (6,5)
- (C) (6,8)
- (D) (4, 10)

Ans: Correct option - A

Because, the maximum of Z occurs at (5, 0).

29. Refer to Exercise 27. (Maximum value of Z+ Minimum value of Z) is equal to

- (A) 13
- (B) 1
- (C)-13
- (D)-17

Ans: Correct option - D

Because, maximum value of Z+ minimum value of Z =15-32=-17

30. The feasible region for an LPP is shown in the Fig. 12.13. Let $F=3x-4y$ be the objective function. Maximum value of F is.

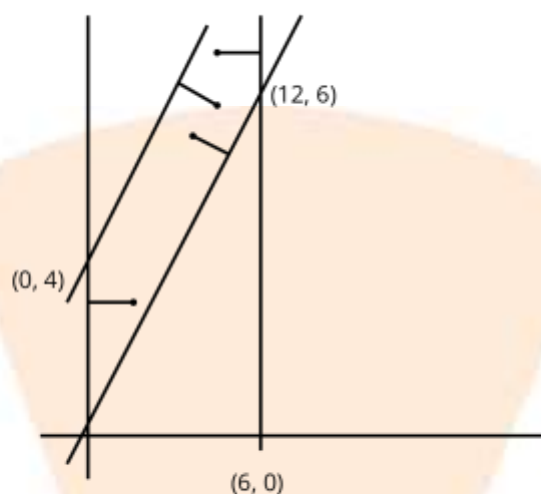


Fig. 12.13

- (A) 0
- (B) 8
- (C) 12
- (D) -18

Ans: Correct option - C

Corner points	Value of
(0,0)	0
(12,6)	12 (maximum)
(0,4)	-16(minimum)

So, the maximum value of F is 12.

31. Refer to Exercise 30. Minimum value of F is

- (A) 0
- (B) -16

(C) 12

(D) does not exist

Ans: Correct option - B

Because, the minimum value of F is -16 at point (0,4).

32. Corner points of the feasible region for an LPP are (0,2), (3,0),(0,5).

Let $F = 4x + 6y$ be the objective function.

The minimum value of F occurs at

(A) (0, 2) only

(B) (3, 0) only

(C) the midpoint of the line segment joining the points (0, 2) and (3, 0) only

(D) any point on the line segment joining the points (0, 2) and (3, 0).

Ans: Correct option - D

Corner points	Value of
(0, 2)	12 (minimum)
(3,0)	12 (maximum)
(6,0)	24
(6,8)	72 (minimum)
(0,5)	30

So, the minimum value of F occurs at any points on the line segment joining the points (0, 2) and (3, 0).

33. Refer to Exercise 32, Maximum of F - Minimum of F is equal to

(A) 60

(B) 48

(C) 42

(D) 18

Ans: Correct option - A

Because, maximum of F - minimum of F = 72 - 12 = 60.

34. Corner points of the feasible region determined by the system of linear constraints are (0,3), (1,1) and (3,0). Let $Z = px + qy$, where $p, q > 0$. Condition on p and q so that the minimum of Z occurs at (3,0) and (1,1) is

(A) $p = 2q$

(B) $p = \frac{q}{2}$

(C) $p = 3q$

(D) $p = q$

Ans: Correct option - B

Corner points	Value of
(0,3)	3q
(1,1)	p+q
(3,0)	3p

So, minimum of z occurs at (3,0) and (1,1) IS

$$p + q = 3p$$

$$\text{Or } 2p = q$$

$$\text{So, } p = \frac{q}{2}$$

Fill in the blanks in each of the Exercises 35 to 41.

35. In a LPP, the linear inequalities or restrictions on the variables are called.....

Ans: In a LPP, the linear inequalities or restrictions on the variables are called linear constraints.

36. In a LPP, the objective function is always.....

Ans: In a LPP, the objective function is always linear.

37. If the feasible region for a LPP is....., then the optimal value of the objective function $Z=ax+by$ may or may not exist.

Ans: If the feasible region for a LPP is unbounded, then the optimal value of objective function $Z=ax+by$ may or may not exist.

38. In a LPP if the objective function $Z=ax+by$ has the same maximum value on two corner points of the feasible region, then every point on the line segment joining these two points give the same.....value.

Ans: In a L.P.P, if the objective function $Z=ax+by$ has the same maximum value on two corner points of the feasible region, then every point on the line segment joining these two points give the same maximum value.

39. A feasible region of a system of linear inequalities is said to be if it can be enclosed within a circle.

Ans: A feasible region of a system of linear inequality is said to be bounded, if it can be enclosed within a circle.

40. A corner point of a feasible region is a point in the region which is theof two boundary lines.

Ans: A corner point of a feasible region is a point in the region which is the intersection of two boundary lines.

41. The feasible region for an LPP is always a.....polygon.

Ans: The feasible region for an LPP is always a convex polygon.

State whether the statements in Exercises 42 to 45 are True or False.

42. If the feasible region for a LPP is unbounded, maximum or minimum of the objective function $Z=ax+by$ may or may not exist.

Ans: True

43. Maximum value of the objective function $Z=ax+by$ in a LPP always occurs at only one corner point of the feasible region.

Ans: False

44. In a LPP, the minimum value of the objective function $Z=ax+by$ is always 0 if origin is one of the corner point of the feasible region.

Ans: False

45. In a LPP, the maximum value of the objective function $Z=ax+by$ is always finite.

Ans: True